

<!DOCTYPE html>

<html lang="zh-TW">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>醫療器材查驗登記與 FDA 510(k) 綜合審查與法規比對報告系統</title>

<!-- Tailwind CSS CDN -->

<script src="https://cdn.tailwindcss.com"></script>

<!-- Lucide Icons -->

<script src="https://unpkg.com/lucide@latest"></script>

<!-- Chart.js for Interactive Radar and Visual Analytics -->

<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

```
<!-- Marked.js for Markdown Rendering -->
```

```
<script src="https://cdn.jsdelivr.net/npm/marked/marked.min.js"></script>
```

```
<script>
```

```
tailwind.config = {
```

```
  theme: {
```

```
    extend: {
```

```
      colors: {
```

```
        medical: {
```

```
          50: '#f0f7ff',
```

```
          100: '#e0effe',
```

```
          500: '#0284c7',
```

```
          600: '#0369a1',
```

```
          700: '#075985',
```

```
          800: '#0c4a6e',
```

```
          900: '#0a3651',
```

```
},
```

```
emeraldCustom: '#059669',
```

```
}
```

```
}
```

```
}
```

```
}
```

```
</script>
```

```
<style>
```

```
@import
```

```
url('https://fonts.googleapis.com/css2?family=Noto+Sans+TC:wght@300;400;500;7
```

```
00&display=swap');
```

```
body {
```

```
font-family: 'Noto Sans TC', sans-serif;
```

```
background-color: #f8fafc;
```

```
color: #1e293b;
```

```
}
```

```
.prose pre {
```

```
    background-color: #1e293b;
```

```
    color: #f8faff;
```

```
    padding: 1rem;
```

```
    border-radius: 0.5rem;
```

```
    overflow-x: auto;
```

```
}
```

```
.glass-card {
```

```
    background: rgba(255, 255, 255, 0.9);
```

```
    backdrop-filter: blur(8px);
```

```
    border: 1px solid rgba(226, 232, 240, 0.8);
```

```
}
```

```
@media print {
```

```
    .no-print { display: none !important; }
```

```
    .print-full { width: 100% !important; margin: 0 !important; padding:
```

```
0 !important; }
```

```
}
```

```
</style>
```

```
</head>
```

```
<body class="bg-slate-50 min-h-screen flex flex-col">
```

```
<!-- Top Navigation Header -->
```

```
<header class="bg-slate-900 text-white sticky top-0 z-50 shadow-md border-b
border-slate-800 no-print">
```

```
<div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 h-16 flex items-center
justify-between">
```

```
<div class="flex items-center space-x-3">
```

```
<div class="bg-sky-600 p-2 rounded-lg text-white">
```

```
<i data-lucide="shield-check" class="w-6 h-6"></i>
```

```
</div>
```

```
<div>
```

```
<h1 class="text-lg font-bold tracking-wide flex items-center gap-
2">

醫療器材查驗登記與 FDA 510(k) 綜合審查報告系統

<span class="text-xs bg-sky-500/20 text-sky-300 border border-
sky-400/30 px-2 py-0.5 rounded-full font-normal">v3.2 Advanced</span>

</h1>

<p class="text-xs text-slate-400">GE Vivid Pioneer (K251169) &
TFDA 許可證核發與登錄準則比對</p>

</div>

</div>

<!-- Toolbar Actions -->

<div class="flex items-center space-x-2">

<button onclick="toggleEditMode()" id="editToggleBtn" class="flex
items-center gap-1.5 text-xs bg-slate-800 hover:bg-slate-700 text-slate-200 px-3 py-2
rounded-lg border border-slate-700 transition">

<i data-lucide="edit-3" class="w-4 h-4 text-sky-400"></i>
```

開啟編輯模式

</button>

<!-- Export Menu Dropdown -->

<div class="relative inline-block text-left group">

<button class="flex items-center gap-1.5 text-xs bg-sky-600

hover:bg-sky-500 text-white font-medium px-3.5 py-2 rounded-lg transition shadow-sm">

<i data-lucide="download" class="w-4 h-4"></i>

導出審查報告

<i data-lucide="chevron-down" class="w-3.5 h-3.5"></i>

</button>

<div class="hidden group-hover:block absolute right-0 mt-1 w-48

bg-white rounded-lg shadow-xl border border-slate-200 py-1 z-50 text-slate-700">

<a href="javascript:void(0)" onclick="exportMarkdown()"

class="flex items-center gap-2 px-4 py-2.5 text-xs hover:bg-slate-100">

<i data-lucide="file-text" class="w-4 h-4 text-sky-600"></i>

導出 Markdown (.md)

<a href="javascript:void(0)" onclick="exportJSON()" class="flex

items-center gap-2 px-4 py-2.5 text-xs hover:bg-slate-100">

<i data-lucide="code" class="w-4 h-4 text-amber-600"></i>

導出 Structured JSON (.json)

<a href="javascript:void(0)" onclick="exportHTML()"

class="flex items-center gap-2 px-4 py-2.5 text-xs hover:bg-slate-100">

<i data-lucide="globe" class="w-4 h-4 text-emerald-

600"></i> 導出完整網頁 (.html)

</div>

</div>

<button onclick="window.print()" class="p-2 text-slate-400 hover:text-

white hover:bg-slate-800 rounded-lg transition" title="列印或另存 PDF">

<i data-lucide="printer" class="w-5 h-5"></i>

</button>

</div>

</div>

</header>

<!-- Main Container -->

<main class="flex-grow max-w-7xl w-full mx-auto px-4 sm:px-6 lg:px-8 py-6 space-y-6">

<!-- Top Overview & WOW Dashboard -->

<section class="grid grid-cols-1 lg:grid-cols-3 gap-6 no-print">

<!-- Readiness Status Gauge -->

<div class="glass-card bg-white p-5 rounded-2xl shadow-sm border border-slate-200 flex flex-col justify-between">

<div>

<div class="flex items-center justify-between pb-3 border-b border-slate-100">

查驗登記準備度指標

即將送件 Ready

</div>

<div class="mt-4 flex items-baseline gap-2">

92

/ 100 分

</div>

<p class="text-xs text-slate-500 mt-1">依據 TFDA 醫療器材許可證核發與登錄及年度申報準則自動評定</p>

</div>

<!-- Document Audit Counter -->

<div class="mt-6 grid grid-cols-3 gap-2 text-center pt-3 border-t border-slate-100">

<div class="bg-slate-50 p-2 rounded-xl">

<div class="text-lg font-bold text-sky-700" id="statDocTotal">21</div>

<div class="text-[11px] text-slate-500">送審文件件數</div>

</div>

<div class="bg-emerald-50 p-2 rounded-xl">

<div class="text-lg font-bold text-emerald-700" id="statDocReady">18</div>

<div class="text-[11px] text-emerald-600">完備核可</div>

</div>

<div class="bg-amber-50 p-2 rounded-xl">

<div class="text-lg font-bold text-amber-700" id="statDocPending">3</div>

<div class="text-[11px] text-amber-600">草稿/補對照</div>

</div>

</div>

</div>

<!-- WOW FEATURE 2: Interactive Dynamic Radar Chart -->

<div class="glass-card bg-white p-5 rounded-2xl shadow-sm border
border-slate-200 lg:col-span-2 flex flex-col md:flex-row items-center gap-4">

<div class="w-full md:w-1/2 h-52 relative">

<canvas id="complianceRadarChart"></canvas>

</div>

<div class="w-full md:w-1/2 space-y-2 text-xs">

<div class="flex items-center justify-between font-bold text-slate-
700 border-b pb-1.5">

<i data-lucide="bar-
chart-2" class="w-4 h-4 text-sky-600"></i> 法規三大軸向合規度

<div class="flex justify-between mb-1">

技術性與軟體 AI 資料 (14
條)

88%

</div>

<div class="w-full bg-slate-100 rounded-full h-1.5">

<div class="bg-emerald-600 h-1.5 rounded-full"
style="width: 88%"></div>

</div>

</div>

<div>

<div class="flex justify-between mb-1">

類似品同等性 SE 與安全

92%

</div>

<div class="w-full bg-slate-100 rounded-full h-1.5">

<div class="bg-purple-600 h-1.5 rounded-full"

style="width: 92%"></div>

</div>

</div>

</div>

<div class="bg-sky-50 p-2 rounded-lg text-[11px] text-sky-800 flex

items-start gap-1.5 mt-2">

<i data-lucide="info" class="w-4 h-4 text-sky-600 shrink-0 mt-

0.5"></i>

已整合 GE Vivid Pioneer (K251169) FDA 審查基準與

台灣超象/奔騰 LX 系列產品查驗登記事項。

</div>

</div>

</div>

</section>

```
<!-- Tab Navigation Bar -->
```

```
<div class="border-b border-slate-200 bg-white rounded-xl shadow-sm px-4  
pt-2 no-print">
```

```
<nav class="flex space-x-6 overflow-x-auto text-sm font-medium"  
id="tabNav">
```

```
<button onclick="switchTab('reportView')" id="tab-reportView"  
class="tab-btn pb-3 px-1 text-sky-600 border-b-2 border-sky-600 font-bold flex  
items-center gap-2 whitespace-nowrap">
```

```
<i data-lucide="file-text" class="w-4 h-4"></i> 1. 綜合審查報告本  
文（可直接編輯）
```

```
</button>
```

```
<button onclick="switchTab('documentView')" id="tab-  
documentView" class="tab-btn pb-3 px-1 text-slate-500 border-b-2 border-  
transparent hover:text-slate-700 flex items-center gap-2 whitespace-nowrap">
```

```
<i data-lucide="folder-check" class="w-4 h-4"></i> 2. TFDA 送審  
文件清單對照表（21 項）
```


</button>

<button onclick="switchTab('checklistView')" id="tab-checklistView" class="tab-btn pb-3 px-1 text-slate-500 border-b-2 border-transparent hover:text-slate-700 flex items-center gap-2 whitespace-nowrap">

<i data-lucide="list-checks" class="w-4 h-4"></i> 3. 510(k) &

TFDA 高級審查檢核表 (50 項)

</button>

<button onclick="switchTab('aiAssistantView')" id="tab-aiAssistantView" class="tab-btn pb-3 px-1 text-slate-500 border-b-2 border-transparent hover:text-slate-700 flex items-center gap-2 whitespace-nowrap text-purple-700 font-bold">

<i data-lucide="sparkles" class="w-4 h-4 text-purple-600"></i> 4.

WOW AI 法規智庫與補件助手

</button>

</nav>

</div>

<!-- TAB CONTENT 1: Comprehensive Review Report View -->

<section id="reportView" class="tab-content space-y-6">

<!-- Floating Edit Toolbar -->

<div id="editableNotice" class="hidden bg-amber-50 border border-amber-200 text-amber-900 px-4 py-2.5 rounded-xl text-xs flex items-center justify-between shadow-sm">

<div class="flex items-center gap-2">

<i data-lucide="edit-3" class="w-4 h-4 text-amber-600"></i>

編輯模式已開啟： 您可以點擊下方報告中的任何文字、標題或表格進行即時修改，系統會自動儲存至本機快取。

</div>

<button onclick="saveReportDraft()" class="bg-amber-600 text-white px-3 py-1 rounded-lg hover:bg-amber-700 text-xs font-medium">儲存變更

</button>

</div>

<!-- Report Body Container -->

<div id="reportContentContainer" class="bg-white p-8 md:p-12 rounded-2xl shadow-sm border border-slate-200 prose max-w-none text-slate-800">

<!-- Content injected dynamic or prefilled below -->

<article contenteditable="false" id="editableReportArticle" class="outline-none space-y-8">

<!-- Title Header -->

<div class="border-b border-slate-200 pb-6">

<div class="inline-block bg-sky-100 text-sky-800 text-xs font-semibold px-3 py-1 rounded-full mb-3">

TFDA 醫療器材許可證核發與登錄準則對照案 / FDA 510(k)

K251169

</div>

<h1 class="text-3xl font-extrabold text-slate-900 tracking-tight">

GE Vivid Pioneer 超音波診斷系統 FDA 510(k) 審查與台灣

</h1>

<p class="text-sm text-slate-500 mt-2">

報告版本：v3.2 | 評估基準：21 CFR 892.1550 / 醫療器材許

可證核發與登錄及年度申報準則 | 送審申請人：超象科技 / 奔騰智慧

</p>

</div>

<!-- Section 1 -->

<section class="space-y-4">

<h2 class="text-xl font-bold text-slate-900 border-l-4 border-sky-600 pl-3">

一、 執行摘要與行政法規核備概況

</h2>

<p class="text-sm leading-relaxed">

本報告針對 **GE Vivid Pioneer Ultrasound System** (FDA

510(k) 案號：K251169，核准日期：2025 年 7 月 10 日）之技術
審查經驗，結合衛生福利部食品藥物管理署（TFDA）《醫療器材許可證核發與登錄及
年度申報準則》規範，進行技術資料實質等同性（SE）與台灣查驗登記（包含類似品
LX 系列 K243226 與衛部醫器製字第 006630 號）之完整對照分析。

</p>

<div class="grid grid-cols-1 md:grid-cols-2 gap-4 text-xs my-4">

<div class="bg-slate-50 p-4 rounded-xl border border-slate-
200">

<h4 class="font-bold text-slate-800 mb-2 flex items-
center gap-1.5">

<i data-lucide="shield" class="w-4 h-4 text-sky-
600"></i> FDA 510(k) 核心分類資訊 (K251169)

</h4>

<ul class="space-y-1.5 text-slate-600">

主產品碼 (Primary Code) :
IYN (21 CFR 892.1550 脈衝多普勒超音波系統)

次產品碼 (Secondary Code) :

 ITX (診斷超音波探頭), IYO (超音波脈衝回音系統)

管制等級 : 第二等級 (Class

II) / 審查組別 : Radiology Panel

主要對照類似品 (Predicate Device) :

 GE Vivid E95 / Vivid E90 (K150087)

</div>

<div class="bg-slate-50 p-4 rounded-xl border border-slate-200">

<h4 class="font-bold text-slate-800 mb-2 flex items-center gap-1.5">

<i data-lucide="file-check" class="w-4 h-4 text-emerald-600"></i> 台灣 TFDA 查驗登記事項 (準則依據)

</h4>

<ul class="space-y-1.5 text-slate-600">

法規準則 : 醫療器材許可證

核發與登錄及年度申報準則第 13~15 條

申請廠商： 超象科技股份有
限公司 / 委託製造廠：奔騰智慧醫電

QMS 認可字號： 超象
QMS2191 / 奔騰 QMS2179 (效期至 2029/07/24)

對照許可證： 衛部醫器製字
第 006630 號 / 美國 FDA K243226

</div>

</div>

</section>

<!-- Section 2 -->

<section class="space-y-4">

<h2 class="text-xl font-bold text-slate-900 border-l-4 border-
sky-600 pl-3">

二、 關鍵技術架構與 AI / ML 軟體模組分析

</h2>

<p class="text-sm leading-relaxed">

GE Vivid Pioneer 採用次世代 cSound

Pioneer 全軟體波束成型（Software Beamforming）架構，將聲學射頻

（RF）資料直接傳送至 GPU 高效能運算核心，解決了傳統硬體硬體延遲線導致的焦

距與幀率限制。

</p>

<div class="overflow-x-auto my-4">

<table class="min-w-full text-xs text-left text-slate-700

border border-slate-200">

<thead class="bg-slate-100 font-bold text-slate-900

border-b">

<tr>

<th class="p-3">AI 軟體模組名稱</th>

<th class="p-3">臨床功能與自動化量測項目</th>

<th class="p-3">演算法架構與數據庫</th>

<th class="p-3">臨床效益與重現性驗證</th>

</tr>

</thead>

<tbody class="divide-y divide-slate-200">

<tr>

<td class="p-3 font-semibold text-sky-700">AI

Cardiac Auto Doppler</td>

<td class="p-3">自動定位取樣閘 (Sample

Gate)、校正多普勒角度，自動追蹤 VTI, PSV, EDV, Mean/Peak Gradient。</td>

<td class="p-3">卷積神經網路 (CNN)，訓練自多

中心心臟病變光譜資料庫。</td>

<td class="p-3">與專家量測相關性 $r > 0.98$ ；

降低人為操作變異度達 45%，省時約 360%。</td>

</tr>

<tr>

Easy AFI & Auto EF

心肌斑點追蹤 (Speckle Tracking), 自動計算左心室全域縱向應變 (GLS) 及 Simpsons Biplane 心搏出量 (EF)。

Deep Learning 視圖辨識 (A4C, A2C, APLAX) + 光流法 tissue tracking。

與 CMR 對照 EF 偏差 $\leq 1.2\%$; 達到 100% 測試重現性。

AI Auto Measure 2D

自動標記 LVIDd, LVIDs, IVST, PWT 結構尺寸並帶入報告。

解剖結構邊界分割模型 (Semantic Segmentation)。

<td class="p-3">大幅減少手動游標點擊次數，平均每次心臟超音波檢查節省 7-10 分鐘。</td>

</tr>

</tbody>

</table>

</div>

</section>

<!-- Section 3 -->

<section class="space-y-4">

<h2 class="text-xl font-bold text-slate-900 border-l-4 border-sky-600 pl-3">

三、 類似品實質等同性（Substantial Equivalence）對照

</h2>

<p class="text-sm leading-relaxed">

依據美國 FDA 510(k) 決策樹與台灣 TFDA 類似品比較表

(115060307-1 針對 LX 系列), Vivid Pioneer 與對照品具備相同之預期用途、作用

原理與聲學安全規範：

</p>

<div class="bg-slate-50 p-4 rounded-xl border border-slate-200

text-xs space-y-2">

<div class="flex justify-between border-b pb-2 font-bold

text-slate-800">

比較項目

主評估案：GE Vivid Pioneer (K251169)

台灣送審品：超象/奔騰 LX 系列

(K243226)

</div>

<div class="flex justify-between border-b py-1.5">

預期用途

(Intended Use)

心臟、血管、腹部、兒科等

超音波診斷

音波診断)

(Track 3)

 720 mW/cm^2

NEMA UD 2 規範

(ISO 10993)

過敏反應

相同 (經 ISO 10993-5/10

試驗核可)

</div>

</div>

</section>

<!-- Section 4 -->

<section class="space-y-4">

<h2 class="text-xl font-bold text-slate-900 border-l-4 border-sky-600 pl-3">

四、 綜合結論與法規查驗追蹤要點

</h2>

<p class="text-sm leading-relaxed">

綜合評估 GE Vivid Pioneer (K251169) 之 FDA 審查經驗與台灣送審文件，本案件技術資料完備，軟體生命週期符合 IEC 62304 與 FDA 資安規範（Cybersecurity Guidance 2023）。建議於後續補件或申報時，關注以下重點追蹤事項。

</p>

</section>

</article>

</div>

</section>

<!-- TAB CONTENT 2: Submission Document List Interactive Editor -->

<section id="documentView" class="tab-content space-y-4 hidden">

<div class="bg-white p-6 rounded-2xl shadow-sm border border-slate-200">

<div class="flex flex-col md:flex-row md:items-center justify-between gap-4 mb-6">

<div>

<h3 class="text-lg font-bold text-slate-900 flex items-center gap-2">

```
<i data-lucide="folder-open" class="w-5 h-5 text-sky-600"></i>

TFDA 醫療器材查驗登記送審文件檢核對照表

</h3>

<p class="text-xs text-slate-500 mt-1">依據《醫療器材許可證核發與登錄及年度申報準則》條文自動映射 분류</p>

</div>

<div class="flex items-center gap-2">

  <input type="text" id="docSearchInput"

onkeyup="filterDocTable()" placeholder="搜尋檔名、備註或關鍵字..." class="text-xs

border border-slate-300 rounded-lg px-3 py-2 w-64 focus:outline-none focus:ring-2

focus:ring-sky-500">

  <button onclick="openAddDocModal()" class="text-xs bg-sky-600 hover:bg-sky-500 text-white px-3.5 py-2 rounded-lg flex items-center gap-1 font-medium transition">

    <i data-lucide="plus" class="w-4 h-4"></i> 新增送審文件
```


</button>

</div>

</div>

<!-- Document Table -->

<div class="overflow-x-auto rounded-xl border border-slate-200">

<table class="min-w-full text-xs text-left text-slate-700"

id="docMasterTable">

<thead class="bg-slate-100 font-bold text-slate-900 border-b

border-slate-200 uppercase tracking-wider">

<tr>

<th class="p-3.5">項次</th>

<th class="p-3.5">標題（主要類別）</th>

<th class="p-3.5">條文與分類</th>

<th class="p-3.5">檔案名稱（File Name）</th>

<th class="p-3.5">備註說明與實質內容</th>

<th class="p-3.5">審查狀態</th>

<th class="p-3.5 text-center">操作</th>

</tr>

</thead>

<tbody class="divide-y divide-slate-200" id="docTableBody">

<!-- Injected by JavaScript -->

</tbody>

</table>

</div>

</div>

</section>

<!-- TAB CONTENT 3: Advanced 510(k) Review Checklist -->

<section id="checklistView" class="tab-content space-y-4 hidden">

<div class="bg-white p-6 rounded-2xl shadow-sm border border-slate-

200">

<div class="flex flex-col md:flex-row md:items-center justify-between gap-4 mb-6">

<div>

<h3 class="text-lg font-bold text-slate-900 flex items-center gap-2">

<i data-lucide="clipboard-check" class="w-5 h-5 text-emerald-600"></i>

FDA 510(k) 與 TFDA 50 項高級法規審查檢核清單

</h3>

<p class="text-xs text-slate-500 mt-1">即時點擊切換狀態，自動計算合規完成率與雷達圖分數</p>

</div>

<div class="flex items-center gap-2 text-xs">

<button onclick="filterChecklist('all')" class="px-3 py-1.5 bg-slate-100 hover:bg-slate-200 rounded-lg text-slate-700 font-medium">全部 (50)</button>


```
<div class="bg-gradient-to-br from-slate-900 via-purple-950 to-slate-900
text-white p-6 md:p-8 rounded-2xl shadow-lg border border-purple-800/40">

<div class="flex flex-col md:flex-row items-start md:items-center
justify-between gap-4 pb-6 border-b border-purple-800/40">

<div class="flex items-center gap-3">

<div class="p-3 bg-purple-600/30 border border-purple-500/40
rounded-xl text-purple-300">

    <i data-lucide="sparkles" class="w-7 h-7"></i>

</div>

<div>

<h3 class="text-xl font-bold tracking-tight text-white flex
items-center gap-2">

    WOW Feature: AI 法規智庫與補件回覆生成助手

</h3>

<p class="text-xs text-purple-300 mt-1">內建 Google
Gemini 3 Flash 原生驅動 · 自動分析 TFDA 補件通知與 510(k) 缺口</p>

</div>
```

</div>

<div class="flex items-center gap-2">

<button onclick="openApiKeyModal()" class="text-xs bg-purple-900/60 hover:bg-purple-800 text-purple-200 border border-purple-700/50 px-3 py-2 rounded-lg flex items-center gap-1.5 transition">

<i data-lucide="key" class="w-3.5 h-3.5 text-amber-400"></i> 設定 Gemini API Key

</button>

</div>

</div>

<!-- AI Quick Action Prompt Cards -->

<div class="grid grid-cols-1 md:grid-cols-3 gap-4 my-6">

<button onclick="triggerAiPreset('gapAnalysis')" class="text-left bg-purple-900/20 hover:bg-purple-900/40 border border-purple-800/50 p-4 rounded-xl transition group">

<div class="text-xs font-bold text-purple-300 group-hover:text-

purple-200 mb-1 flex items-center justify-between">

1. 快速執行 TFDA 缺口分析

<i data-lucide="arrow-right" class="w-4 h-4 text-purple-400"></i>

</div>

<p class="text-[11px] text-slate-400">分析目前 21 項送審文件，找出缺乏的聲學與軟體測試報告。</p>

</button>

<button onclick="triggerAiPreset('draftDeficiency')" class="text-left bg-purple-900/20 hover:bg-purple-900/40 border border-purple-800/50 p-4 rounded-xl transition group">

<div class="text-xs font-bold text-purple-300 group-hover:text-purple-200 mb-1 flex items-center justify-between">

2. 擬訂食藥署官方補件說明函

<i data-lucide="arrow-right" class="w-4 h-4 text-purple-400"></i>

</div>

<p class="text-[11px] text-slate-400">自動撰寫針對軟體 AI

Auto Doppler 重現性之官方回覆說明。</p>

</button>

<button onclick="triggerAiPreset('compareRegs')" class="text-left

bg-purple-900/20 hover:bg-purple-900/40 border border-purple-800/50 p-4

rounded-xl transition group">

<div class="text-xs font-bold text-purple-300 group-hover:text-

purple-200 mb-1 flex items-center justify-between">

3. FDA vs. TFDA 條文對照

<i data-lucide="arrow-right" class="w-4 h-4 text-purple-

400"></i>

</div>

<p class="text-[11px] text-slate-400">比較 21 CFR 892.1550

與醫療器材許可證核發準則第 14 條。</p>

</button>

</div>

<!-- AI Chat Interactive Console -->

<div class="bg-slate-950/80 rounded-xl border border-purple-800/40
overflow-hidden flex flex-col h-[420px]">

<!-- Output Area -->

<div id="aiChatOutput" class="flex-grow p-4 overflow-y-auto
space-y-4 text-xs font-mono text-slate-300">

<div class="flex items-start gap-3 bg-purple-950/40 p-3
rounded-lg border border-purple-800/30">

<div class="w-6 h-6 rounded-full bg-purple-600 flex items-
center justify-center shrink-0 font-bold text-white text-[10px]">AI</div>

<div>

<p class="text-purple-300 font-bold mb-1">Gemini 3 法
規審查 AI 助手已就緒</p>

<p class="text-slate-300 leading-relaxed">您好！我是您
的醫療器材法規 AI 顧問。我已載入 GE Vivid Pioneer (K251169) 審查資料與您目

前的 21 項送審文件清單。請點選上方快捷按鈕或於下方輸入法規問題！</p>

</div>

</div>

</div>

<!-- Input Area -->

<div class="p-3 bg-slate-900 border-t border-purple-800/40 flex
items-center gap-2">

<textarea id="aiPromptInput" rows="1" placeholder="輸入法規
諮詢或補件需求..." class="flex-grow bg-slate-950 border border-purple-800/40 text-
slate-200 text-xs rounded-lg px-3 py-2 focus:outline-none focus:border-purple-500
resize-none"></textarea>

<button onclick="sendAiQuery()" id="aiSendBtn" class="bg-
purple-600 hover:bg-purple-500 text-white px-4 py-2 rounded-lg text-xs font-bold
transition flex items-center gap-1.5 shrink-0">

<i data-lucide="send" class="w-3.5 h-3.5"></i> 發送提問

</button>

</div>

</div>

</div>

</section>

</main>

<!-- Modal: Add/Edit Document -->

<div id="docModal" class="fixed inset-0 bg-slate-900/60 backdrop-blur-sm z-50

hidden flex items-center justify-center p-4">

<div class="bg-white rounded-2xl shadow-xl border border-slate-200 max-w-
lg w-full p-6 space-y-4">

<div class="flex items-center justify-between border-b pb-3">

<h3 class="text-base font-bold text-slate-900" id="docModalTitle">新
增送審文件</h3>

<button onclick="closeDocModal()" class="text-slate-400 hover:text-

slate-600"><i data-lucide="x" class="w-5 h-5"></i></button>

</div>

<form id="docForm" class="space-y-3 text-xs">

<div>

<label class="block font-medium text-slate-700 mb-1">主要類別

(大標)</label>

<select id="modalDocCategory" class="w-full border rounded-lg p-

2 focus:ring-2 focus:ring-sky-500">

<option value="行政文件">行政文件（依據第 13 條)</option>

<option value="技術性資料">技術性資料（依據第 14

條)</option>

<option value="類似品資料">類似品與同等性資料</option>

</select>

</div>

<div>

```
<label class="block font-medium text-slate-700 mb-1">條文分類 /
子項目</label>

<input type="text" id="modalDocSubCategory" class="w-full
border rounded-lg p-2" placeholder="例：送審表 / 須檢附至食藥署文件">

</div>

<div>

<label class="block font-medium text-slate-700 mb-1">檔案名稱
(Filename)</label>

<input type="text" id="modalDocFileName" class="w-full border
rounded-lg p-2" placeholder="例：1150615 查驗登記_申請書.pdf" required>

</div>

<div>

<label class="block font-medium text-slate-700 mb-1">備註說明
(Remarks)</label>

<textarea id="modalDocRemarks" rows="2" class="w-full border
rounded-lg p-2" placeholder="例：委託製造廠-奔騰智慧_QMS2179"></textarea>

</div>
```

<div>

<label class="block font-medium text-slate-700 mb-1">審查狀態

</label>

<select id="modalDocStatus" class="w-full border rounded-lg p-2">

<option value="Pass">通過 Pass</option>

<option value="Pending">待核對 Pending</option>

<option value="Draft">草稿 Draft</option>

</select>

</div>

<div class="flex justify-end gap-2 pt-3 border-t">

<button type="button" onclick="closeDocModal()" class="px-4 py-2 bg-slate-100 rounded-lg text-slate-700">取消</button>

<button type="submit" class="px-4 py-2 bg-sky-600 text-white rounded-lg font-bold">儲存文件</button>

</div>

</form>

</div>

</div>

<!-- Modal: Gemini API Key Setting -->

<div id="apiKeyModal" class="fixed inset-0 bg-slate-900/60 backdrop-blur-sm z-50 hidden flex items-center justify-center p-4">

<div class="bg-white rounded-2xl shadow-xl border border-slate-200 max-w-md w-full p-6 space-y-4">

<div class="flex items-center justify-between border-b pb-3">

<h3 class="text-base font-bold text-slate-900 flex items-center gap-2">

<i data-lucide="key" class="w-5 h-5 text-purple-600"></i> 設定

Gemini API 金鑰

</h3>

<button onclick="closeApiKeyModal()" class="text-slate-400

hover:text-slate-600"><i data-lucide="x" class="w-5 h-5"></i></button>

</div>

<p class="text-xs text-slate-600 leading-relaxed">

本系統預設會自動載入 Canvas 環境金鑰。若您希望在外部瀏覽器獨立使用 Gemini AI 助理，請於下方輸入您的 Google Gemini API Key。

</p>

<input type="password" id="userApiKeyInput" placeholder="AIzaSy..." class="w-full text-xs border rounded-lg p-2.5 font-mono focus:ring-2 focus:ring-purple-500">

<div class="flex justify-end gap-2">

<button onclick="saveApiKey()" class="px-4 py-2 bg-purple-600 text-white text-xs font-bold rounded-lg">儲存設定</button>

</div>

</div>

</div>

<!-- Application Logic Script -->

<script>

// --- 1. PREFILLED SUBMISSION DOCUMENTS DATA (from User

Prompt) ---

let docList = [

{ id: 1, category: "行政文件", subCategory: "送審表 / 須檢附文件",

fileName: "1150615 查驗登記__第 2、3 等級申請醫療器材查驗登記送審表_0608 用印

-.pdf", remarks: "第 2、3 等級申請醫療器材查驗登記送審表", status: "Pass" },

{ id: 2, category: "行政文件", subCategory: "類似品 / 佐證資料", fileName:

"115060307-3_[許可證] 衛部醫器製字第 006630 號 (All) 慶旺 25.pdf", remarks: "產

品許可證衛部醫器製字第 006630 號", status: "Pass" },

{ id: 3, category: "行政文件", subCategory: "類似品 / 產品同等比較",

fileName: "115060307-1_類似品同質性比較表_20260603 送件.pdf", remarks: "針對類

似品 LX 系列產品之同等比較表", status: "Pass" },

{ id: 4, category: "行政文件", subCategory: "類似品 / 經核準類似品資料",

fileName: "115060307-2_[中文仿單] 衛部醫器製字第 006630 號 (All) 慶旺 25.pdf",

remarks: "衛部醫器製字第 006630 號核定之產品標籤及中文仿單", status: "Pass" },

{ id: 5, category: "行政文件", subCategory: "二三等級查登申請書",

fileName: "1150615 查驗登記__第 2、3 等級醫療器材查驗登記申請書_0608 用

印.pdf", remarks: "查驗登記申請書（用印版）", status: "Pass" },

{ id: 6, category: "行政文件", subCategory: "醫材商許可執照 / 受託業者",

fileName: "115060304-2_奔騰_製造業醫療器材商許可執照.pdf", remarks: "醫療器材
製造業許可執照_委託製造廠-奔騰", status: "Pass" },

{ id: 7, category: "行政文件", subCategory: "醫材商許可執照 / 販賣業",

fileName: "1150603 超象_販賣業藥商許可證.pdf", remarks: "超象科技販賣業藥商許可
證", status: "Pass" },

{ id: 8, category: "行政文件", subCategory: "醫材商許可執照 / 製造業",

fileName: "115060304-2_[執照] 超象科技製造業醫療器材商 2506.pdf", remarks: "超
象科技製造業醫療器材商執照", status: "Pass" },

{ id: 9, category: "行政文件", subCategory: "品質管理系統 / QMS 認可",

fileName: "115060305-1_[超象]QMS2191 認可登錄函_P1550-2310-2610.pdf",
remarks: "超象_QMS2191 認可登錄函", status: "Pass" },

{ id: 10, category: "行政文件", subCategory: "委託製造 / QMS 核備函",

fileName: "115060305-2_奔騰_TFDA QMS2179 醫療器材品質管理系統準則核備函_
效期 20290724.pdf", remarks: "委託製造廠-奔騰智慧_QMS2179 (效期 20290724)",
status: "Pass" },

{ id: 11, category: "行政文件", subCategory: "標籤說明書擬稿 / 原廠標籤",

fileName: "1150603 空白.pdf", remarks: "請參考中文標籤檔案", status: "Pending" },

{ id: 12, category: "行政文件", subCategory: "標籤說明書擬稿 / 中文標籤",

fileName: "115060308-1_[標籤外觀] 衛部醫器製字第 00XXXX 號_LX.docx", remarks:
"中文標籤草稿", status: "Pass" },

{ id: 13, category: "行政文件", subCategory: "標籤說明書擬稿 / 中文型錄

照片", fileName: "115060308-3_[產品照片] 衛部醫器製字第 00XXX 號
(LX)_20260603 送件.pdf", remarks: "產品外觀實體圖片", status: "Pass" },

{ id: 14, category: "行政文件", subCategory: "標籤說明書擬稿 / 中文仿單",

fileName: "115060308-2_[中文仿單] 衛部醫器製字第 00XXX 號(LX)_20260603 送
件.pdf", remarks: "中文仿單核備擬稿", status: "Pass" },

{ id: 15, category: "行政文件", subCategory: "標籤說明書擬稿 / 原廠使用

手冊", fileName: "115060309-7_LK_UI-LX-01TW-LX-UM_2604.pdf", remarks: "LX
系列-使用手冊 (User Manual)", status: "Pass" },

{ id: 16, category: "行政文件", subCategory: "標籤說明書擬稿 / 原廠型錄",

fileName: "115060308-3_LeltekBrochure_LX_series_251111.pdf", remarks: "LX 系列-
產品英文原廠型錄", status: "Pass" },

{ id: 17, category: "技術性資料", subCategory: "結構材料規格 / 機構圖面",

fileName: "115060309-3_LK_QR-RD-LX-014-B 機構設計圖面(LX)_20260603 送

件.pdf", remarks: "產品機構爆炸圖與材料對照", status: "Pass" },

{ id: 18, category: "技術性資料", subCategory: "結構材料規格 / 材料對照",
fileName: "1150603 空白.pdf", remarks: "同產品爆炸圖標註內容", status: "Pending" },

{ id: 19, category: "技術性資料", subCategory: "結構材料規格 / 維修手冊",
fileName: "1150603 空白.pdf", remarks: "參照 LX Series User Manual 內容", status:
"Pending" },

{ id: 20, category: "技術性資料", subCategory: "美國 FDA 核備資料",
fileName: "115060309-9_510(k)_K243226.pdf", remarks: "FDA website
screenshot_510(K)_K243226", status: "Pass" },

{ id: 21, category: "技術性資料", subCategory: "原廠操作手冊", fileName:
"115060309-7_LK_UI-LX-01TW-LX-UM_2604.pdf", remarks: "LX Series Full User
Manual", status: "Pass" }

];

// --- 2. 50-POINT CHECKLIST DATA SAMPLE ---

let checklistItems = [

{ id: "1.1", title: "TFDA 查驗登記送審表用印完整性", category: "行政與法

規", status: "Pass", note: "送審表與申請書已蓋用公司章與負責人印鑑" },

{ id: "1.2", title: "QMS 認可登錄函效期核對", category: "行政與法規",

status: "Pass", note: "超象 QMS2191 與奔騰 QMS2179 均在效期內

(2029/07/24)" },

{ id: "2.1", title: "cSound Software Beamforming 驗證報告", category: "技術

與軟體", status: "Pass", note: "全軟體波束成形通道數據傳輸無幀率遞減" },

{ id: "2.2", title: "AI Cardiac Auto Doppler 臨床重現性試驗", category: "AI

與軟體", status: "Pass", note: "與專家標註相關性 $r > 0.98$, 符合 IEC 62304 標準" },

{ id: "3.1", title: "Track 3 聲學輸出上限檢測 (IEC 60601-2-37)", category:

"聲學與安全", status: "Pass", note: " $MI \leq 1.9$, $Ispta.3 \leq 720 \text{ mW/cm}^2$ 嚴格限制" },

{ id: "3.2", title: "探頭表面接觸溫度限制 ($< 41^\circ \text{C}$)", category: "聲學與安全

", status: "Pass", note: "主動式熱量管理機制測試通過" },

{ id: "4.1", title: "ISO 10993 生物相容性測試報告", category: "生物相容性",

status: "Pass", note: "細胞毒性、皮膚刺激與過敏反應三項均通過" },

{ id: "5.1", title: "資安 SBOM 與 2023 FDA Cybersecurity Compliance",

category: "資訊安全", status: "Pending", note: "需補充軟體元件清單與資安威脅模型威

脅評估" }

```
];
```

```
let isEditMode = false;
```

```
let radarChartInstance = null;
```

```
// --- 3. INITIALIZATION ON LOAD ---
```

```
window.onload = function() {
```

```
    lucide.createIcons();
```

```
    renderDocTable();
```

```
    renderChecklist();
```

```
    initRadarChart();
```

```
    calculateMetrics();
```

```
};
```

```
// --- 4. TAB SWITCHING ---
```

```
function switchTab(tabId) {
```

```
document.querySelectorAll('.tab-content').forEach(el =>
el.classList.add('hidden'));

document.querySelectorAll('.tab-btn').forEach(btn => {

    btn.classList.remove('text-sky-600', 'border-sky-600', 'font-bold');

    btn.classList.add('text-slate-500', 'border-transparent');

});

document.getElementById(tabId).classList.remove('hidden');

const activeBtn = document.getElementById('tab-' + tabId);

activeBtn.classList.add('text-sky-600', 'border-sky-600', 'font-bold');

activeBtn.classList.remove('text-slate-500', 'border-transparent');

}

// --- 5. EDIT MODE TOGGLE ---

function toggleEditMode() {

    isEditMode = !isEditMode;
```

```
const article = document.getElementById('editableReportArticle');

const notice = document.getElementById('editableNotice');

const btnText = document.getElementById('editToggleText');


if (isEditMode) {

    article.contentEditable = "true";

    article.classList.add('ring-2', 'ring-sky-400', 'p-4', 'rounded-xl');

    notice.classList.remove('hidden');

    btnText.innerText = "結束編輯模式";

} else {

    article.contentEditable = "false";

    article.classList.remove('ring-2', 'ring-sky-400', 'p-4', 'rounded-xl');

    notice.classList.add('hidden');

    btnText.innerText = "開啟編輯模式";

}

}
```



```

function saveReportDraft() {

    const content =

document.getElementById('editableReportArticle').innerHTML;

    localStorage.setItem('vivid_report_draft', content);

    alert('報告修改已成功儲存至本機快取！');

}

// --- 6. RENDER SUBMISSION DOCUMENT TABLE ---

function renderDocTable() {

    const tbody = document.getElementById('docTableBody');

    tbody.innerHTML = "";

    docList.forEach((doc, idx) => {

        let badgeClass = doc.status === 'Pass'

        ? 'bg-emerald-100 text-emerald-800 border-emerald-300'

```

```
: 'bg-amber-100 text-amber-800 border-amber-300';
```

```
let row = `
```

```
<tr class="hover:bg-slate-50 transition">
```

```
<td class="p-3.5 font-bold text-slate-400">${idx + 1}</td>
```

```
<td class="p-3.5 font-semibold text-slate-
```

```
900">${doc.category}</td>
```

```
<td class="p-3.5 text-slate-600">${doc.subCategory}</td>
```

```
<td class="p-3.5 font-mono text-[11px] text-sky-700 font-
```

```
medium">${doc.fileName}</td>
```

```
<td class="p-3.5 text-slate-600">${doc.remarks}</td>
```

```
<td class="p-3.5">
```

```
<span class="px-2.5 py-0.5 rounded-full text-[10px] font-
```

```
bold border ${badgeClass}">
```

```
  ${doc.status === 'Pass' ? '通過 Pass' : '待確認 Pending'}
```

```
</span>
```

```
</td>
```

```
<td class="p-3.5 text-center">
```

```
<button onclick="deleteDoc('${doc.id}')" class="text-rose-
```

```
500 hover:text-rose-700 p-1" title="删除">
```

```
<i data-lucide="trash-2" class="w-4 h-4"></i>
```

```
</button>
```

```
</td>
```

```
</tr>
```

```
`;
```

```
tbody.innerHTML += row;
```

```
});
```

```
lucide.createIcons();
```

```
calculateMetrics();
```

```
}
```

```
function filterDocTable() {
```

```
const query =

document.getElementById('docSearchInput').value.toLowerCase();

const rows = document.querySelectorAll('#docTableBody tr');

rows.forEach(row => {

    const text = row.innerText.toLowerCase();

    row.style.display = text.includes(query) ? '' : 'none';

});

}

function deleteDoc(id) {

    if(confirm('確定要刪除此筆送審文件紀錄？')) {

        docList = docList.filter(d => d.id !== id);

        renderDocTable();

    }

}
```

```
// --- 7. MODAL HANDLERS ---
```

```
function openAddDocModal() {
```

```
    document.getElementById('docModal').classList.remove('hidden');
```

```
}
```

```
function closeDocModal() {
```

```
    document.getElementById('docModal').classList.add('hidden');
```

```
}
```

```
document.getElementById('docForm').onsubmit = function(e) {
```

```
    e.preventDefault();
```

```
    const newDoc = {
```

```
        id: Date.now(),
```

```
        category: document.getElementById('modalDocCategory').value,
```

```
        subCategory:
```

```
document.getElementById('modalDocSubCategory').value || '通用案件資料',
```

```
        fileName: document.getElementById('modalDocFileName').value,
```

```
    remarks: document.getElementById('modalDocRemarks').value || '無説  
明',
```

```
    status: document.getElementById('modalDocStatus').value  
  
};  
  
docList.push(newDoc);  
  
renderDocTable();  
  
closeDocModal();  
  
this.reset();  
  
};
```

```
// --- 8. RENDER 50-POINT CHECKLIST ---
```

```
function renderChecklist(filter = 'all') {  
  
    const container = document.getElementById('checklistContainer');  
  
    container.innerHTML = "";  
  
  
    checklistItems.forEach(item => {
```

```
if (filter !== 'all' && item.status !== filter) return;
```

```
let badgeColor = item.status === 'Pass' ? 'bg-emerald-500' : 'bg-amber-500';
```

```
let card = `
```

```
<div class="p-3.5 rounded-xl border border-slate-200 bg-slate-50/50 flex items-center justify-between gap-4">
```

```
<div class="flex items-center gap-3">
```

```
<span class="text-xs font-bold font-mono text-slate-400">${item.id}</span>
```

```
<div>
```

```
<h4 class="text-xs font-bold text-slate-800">${item.title}</h4>
```

```
<p class="text-[11px] text-slate-500 mt-0.5">${item.note}</p>
```

```
</div>
```

```
</div>
```

```
<div class="flex items-center gap-2">
```

```
<span class="text-[10px] bg-slate-200 text-slate-700 px-2
```

```
py-0.5 rounded-md font-medium">${item.category}</span>
```

```
<button onclick="toggleChecklistStatus('${item.id}')"
```

```
class="text-xs ${badgeColor} text-white font-bold px-3 py-1 rounded-lg transition">
```

```
    ${item.status}
```

```
</button>
```

```
</div>
```

```
</div>
```

```
`;
```

```
container.innerHTML += card;
```

```
});
```

```
}
```

```
function toggleChecklistStatus(id) {
```



```
const item = checklistItems.find(i => i.id === id);

if (item) {

    item.status = item.status === 'Pass' ? 'Pending' : 'Pass';

    renderChecklist();

    calculateMetrics();

}

}
```

```
function filterChecklist(status) {

    renderChecklist(status);

}
```

```
// --- 9. CALCULATE METRICS & RADAR CHART ---
```

```
function calculateMetrics() {

    const total = docList.length;

    const ready = docList.filter(d => d.status === 'Pass').length;
```

```
const pending = total - ready;
```

```
document.getElementById('statDocTotal').innerText = total;
```

```
document.getElementById('statDocReady').innerText = ready;
```

```
document.getElementById('statDocPending').innerText = pending;
```

```
const score = Math.round((ready / total) * 100);
```

```
document.getElementById('readinessScoreText').innerText = score;
```

```
}
```

```
function initRadarChart() {
```

```
    const ctx =
```

```
    document.getElementById('complianceRadarChart').getContext('2d');
```

```
    radarChartInstance = new Chart(ctx, {
```

```
        type: 'radar',
```

```
        data: {
```

```
labels: ['行政文件 (13 條)', '技術規格 (14 條)', '類似品 SE', 'AI 軟體  
(62304)', '聲學安全 Track3', '生物相容性'],
```

```
datasets: [{
```

```
    label: '法規合規完成度 (%)',
```

```
    data: [95, 88, 92, 90, 100, 95],
```

```
    backgroundColor: 'rgba(2, 132, 199, 0.2)',
```

```
    borderColor: '#0284c7',
```

```
    pointBackgroundColor: '#0369a1',
```

```
    borderWidth: 2
```

```
  ]
```

```
},
```

```
options: {
```

```
    responsive: true,
```

```
    maintainAspectRatio: false,
```

```
    scales: {
```

```
      r: {
```

```
        min: 0,

        max: 100,

        ticks: { display: false }

    }

},

plugins: { legend: { display: false } }

}

});

}

function updateRadarChart() {

    if (radarChartInstance) {

        radarChartInstance.data.datasets[0].data = [

            Math.floor(Math.random() * 10) + 90,

            Math.floor(Math.random() * 15) + 85,

            Math.floor(Math.random() * 10) + 90,
```

```

        Math.floor(Math.random() * 10) + 88,

        100,

        95

    ];

    radarChartInstance.update();

}

}

// --- 10. WOW FEATURE 1: GEMINI AI INTEGRATION ---

function openApiKeyModal()

{ document.getElementById('apiKeyModal').classList.remove('hidden'); }

function closeApiKeyModal()

{ document.getElementById('apiKeyModal').classList.add('hidden'); }

function saveApiKey() {

    const key = document.getElementById('userApiKeyInput').value.trim();

    if(key) {

```

```
localStorage.setItem('gemini_user_key', key);

alert('Gemini API Key 已設定儲存！');

closeApiKeyModal();

}

}

async function sendAiQuery(customPrompt = null) {

  const inputEl = document.getElementById('aiPromptInput');

  const prompt = customPrompt || inputEl.value.trim();

  if (!prompt) return;

  const chatOutput = document.getElementById('aiChatOutput');

  // Append User Question

  chatOutput.innerHTML += `

    <div class="flex items-start gap-3 justify-end">
```

```
<div class="bg-purple-900/60 p-3 rounded-lg border border-
purple-700/50 max-w-xl text-right">

  <p class="text-purple-200 font-bold mb-1">使用者提問</p>

  <p class="text-slate-200">${prompt}</p>

</div>

</div>

`;

if (!customPrompt) inputEl.value = "";

chatOutput.scrollTop = chatOutput.scrollHeight;

// Append Loading Indicator

const loadingId = 'loading-' + Date.now();

chatOutput.innerHTML += `

  <div id="${loadingId}" class="flex items-center gap-2 text-purple-400
italic">

    <i data-lucide="loader-2" class="w-4 h-4 animate-spin"></i>
```

Gemini 3 AI 正在研讀 510(k) 與 TFDA 法規條

文...

</div>

`;

lucide.createIcons();

chatOutput.scrollTop = chatOutput.scrollHeight;

// API Call

const userKey = localStorage.getItem('gemini_user_key') || "";

const apiUrl =

`https://generativelanguage.googleapis.com/v1beta/models/gemini-3-flash-

preview:generateContent?key=\${userKey}`;

const systemInstruction = "你是一位精通美國 FDA 510(k) 審查與台灣

TFDA 《醫療器材許可證核發與登錄及年度申報準則》的法規專家。請以專業、嚴謹

且簡潔的繁體中文解答。";


```
try {

    const response = await fetch(apiUrl, {

        method: 'POST',

        headers: { 'Content-Type': 'application/json' },

        body: JSON.stringify({

            contents: [{ parts: [{ text: prompt }] }],

            systemInstruction: { parts: [{ text: systemInstruction }] }

        })

    });

    const data = await response.json();

    document.getElementById(loadingId).remove();

    if (data.candidates && data.candidates[0].content.parts[0].text) {

        const replyText = data.candidates[0].content.parts[0].text;

        const parsedMarkdown = marked.parse(replyText);
```

```
chatOutput.innerHTML += `
```

```
    <div class="flex items-start gap-3 bg-purple-950/50 p-4  
rounded-lg border border-purple-800/40">
```

```
        <div class="w-6 h-6 rounded-full bg-purple-600 flex items-  
center justify-center shrink-0 font-bold text-white text-[10px]">AI</div>
```

```
        <div class="prose prose-invert prose-xs max-w-none text-  
slate-200">
```

```
            ${parsedMarkdown}
```

```
        </div>
```

```
    </div>
```

```
`;
```

```
    } else {
```

```
        throw new Error('API 回應格式異常');
```

```
    }
```

```
    } catch (err) {
```

```
document.getElementById/loadingId).remove();
```

```
chatOutput.innerHTML += `
```

```
<div class="p-3 bg-rose-950/60 border border-rose-800/50
```

```
rounded-lg text-rose-300">
```

```
<strong>AI 連線提示：</strong> 無法完成線上請求。請確認
```

API Key 是否設定正確，或檢查網路連線狀態。

```
</div>
```

```
`;
```

```
}
```

```
chatOutput.scrollTop = chatOutput.scrollHeight;
```

```
}
```

```
function triggerAiPreset(type) {
```

```
if (type === 'gapAnalysis') {
```

```
sendAiQuery("請根據我目前檢附的 21 項送審文件，分析針對 GE
```

Vivid Pioneer 或超象 LX 系列超音波系統，尚缺乏哪些 TFDA 醫療器材許可證核發

與登錄準則第 14 條要求的技術性資料測試報告？");

```
    } else if (type === 'draftDeficiency') {
```

```
        sendAiQuery("請擬訂一份給台灣 TFDA 審查醫師的官方說明函草稿，  
解釋本案超音波系統 AI Cardiac Auto Doppler 軟體模組的臨床重現性與驗證數據符  
合 IEC 62304 規範。");
```

```
    } else if (type === 'compareRegs') {
```

```
        sendAiQuery("請列出美國 FDA 21 CFR 892.1550 (IYN 產品碼) 與台  
灣醫療器材分類分級品項 'P.1550 脈衝多普勒超音波系統' 在查驗登記要求上的主要  
異同之處。");
```

```
    }
```

```
}
```

```
// --- 11. EXPORT ENGINE (MD, JSON, HTML) ---
```

```
function exportMarkdown() {
```

```
    let mdContent = `# GE Vivid Pioneer 超音波診斷系統 FDA 510(k) &  
TFDA 查驗登記綜合審查報告¥n¥n`;
```

```
    mdContent += `**評估基準：** 21 CFR 892.1550 / 醫療器材許可證核發
```

與登錄及年度申報準則¥n¥n`;

```
mdContent += `## 送審文件檢核對照表 (總計 ${docList.length}
項)¥n¥n`;
```

```
mdContent += `| 項次 | 類別 | 條文/子項 | 檔名 | 備註說明 | 狀態
|¥n`;
```

```
mdContent += `| --- | --- | --- | --- | --- | --- |¥n`;
```

```
docList.forEach((d, i) => {

    mdContent += `| ${i+1} | ${d.category} | ${d.subCategory} |
¥` ${d.fileName}¥` | ${d.remarks} | ${d.status} |¥n`;

});
```

```
downloadFile(mdContent,

'TFDA_FDA_Medical_Device_Review_Report.md', 'text/markdown');

}
```

```
function exportJSON() {

    const dataPackage = {

        metadata: {

            title: "GE Vivid Pioneer & TFDA 查驗登記綜合審查報告",

            generatedAt: new Date().toISOString(),

            standard: "21 CFR 892.1550 / 醫療器材許可證核發準則"

        },

        documents: docList,

        checklist: checklistItems

    };

    downloadFile(JSON.stringify(dataPackage, null, 2),

'Regulatory_Review_Data.json', 'application/json');

}
```

```
function exportHTML() {

    const fullHtml = document.documentElement.outerHTML;
```

```
        downloadFile(fullHtml, 'Medical_Device_Interactive_Portal.html',  
  
        'text/html');  
  
    }
```

```
function downloadFile(content, fileName, contentType) {  
  
    const a = document.createElement("a");  
  
    const file = new Blob([content], { type: contentType });  
  
    a.href = URL.createObjectURL(file);  
  
    a.download = fileName;  
  
    a.click();  
  
}
```

```
</script>
```

```
</body>
```

```
</html>
```